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This project has been a great opportunity to apply what I learned in class to a real-world problem. Many people encouraged and supported me along the way, and I truly appreciate everyone who helped make this assignment successful.

Executive Summary

This report presents the findings from an optimized Random Forest model that achieved 87.4% accuracy in predicting customer churn for a banking dataset. Key insights reveal that older customers and those with lower account balances are more likely to churn, while active product engagement and diverse account holdings indicate stronger customer retention. Based on these insights, the report recommends that banks implement targeted retention campaigns for high-risk customer segments, allocate resources to prioritize at-risk high-value customers, develop new products and services aligned with customer needs, and personalize interactions to improve engagement and loyalty. By adopting a dual-model approach using both the powerful Random Forest and the interpretable Logistic Regression, banks can leverage predictive power and clear business intelligence to effectively reduce churn and enhance overall customer satisfaction.

Introduction

The banking industry is highly competitive and keeping existing customers (customer retention) is very important for maintaining profitability. Banks face a significant risk of customers switching to other banks (customer churn) due to factors like low engagement, dissatisfaction, or financial instability. Understanding what predicts whether a customer will churn allows banks to proactively implement targeted retention strategies, improving customer loyalty and the lifetime value of each customer.

This report analyzes customer churn behavior using a dataset of 10,000 banking customers from Kaggle. The analysis uses advanced machine learning techniques to uncover insights into customer behaviors and key factors that contribute to churn. Two different models are used - Random Forest Classifier and Logistic Regression. Random Forest is good at capturing complex, non-linear relationships in the data, while Logistic Regression provides more interpretable insights into the specific drivers of churn. The goal is to help banks better understand and predict customer churn so they can take proactive steps to retain more of their existing customers.

01. Reasons for Dataset Selection

The selected dataset, **Churn_Modelling.csv** from **Kaggle**, contains **10,000 records** and **14 features**, making it ideal for a predictive classification task focused on customer attrition in the banking sector. Each record represents a unique customer, detailing demographic and financial attributes including **"CreditScore," "Age," "Balance,"** and **"NumOfProducts"** which can **influence a customer's likelihood of leaving the bank** (Target variable = **"Exited"**).

(<https://www.kaggle.com/datasets/shrutimechlearn/churn-modelling/data>)

This dataset aligns closely with real-world business challenges in the finance industry, where maintaining competitiveness and profitability relies on understanding and reducing customer churn. By identifying key drivers of attrition, such as low engagement or low creditworthiness, financial institutions can take **targeted actions to retain customers, thereby improving long-term profitability and customer loyalty.**

Predicting customer churn is particularly important for banks because keeping current customers is typically less expensive than finding new ones. High churn rates have a direct impact on profitability because they may be a sign of unmet customer needs or service dissatisfaction. By identifying key indicators of attrition, such as **low credit scores, short tenure, or low product engagement, banks can strategically focus on at-risk customers with personalized retention offers.**

With a balanced combination of continuous and categorical variables to enable precise predictive modelling, this dataset is perfect for **Classification modelling**. Banks can reduce customer loss, improve customer relationships, and eventually increase total customer lifetime value by using the insights produced to guide targeted retention strategies.

Import Necessary Libraries

```
[196]: # Import libraries for data manipulation and visualization
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.gridspec import GridSpec

# Import libraries for preprocessing and model evaluation
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report
from sklearn.model_selection import train_test_split, cross_val_score, cross_val_predict, GridSearchCV, RandomizedSearchCV

# Import libraries for handling imbalanced datasets
from imblearn.over_sampling import SMOTE

# Import machine learning models
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import GaussianNB

# Statistical distributions for hyperparameter tuning
from scipy.stats import randint

# For synthetic dataset creation
from sklearn.datasets import make_classification

[13]: # Drop unnecessary columns to streamline the DataFrame
row_data = row_data.drop(['RowNumber', 'CustomerId', 'Surname'], axis=1)
```

Figure 1 - Import Libraries

	Creditscore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCard	IsActiveMember	EstimatedSalary	Exited
0	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

▼ In this section, unnecessary columns are removed from the DataFrame to streamline its structure. The RowNumber, CustomerId, and Surname columns are dropped, as they may not be essential for the analysis. The resulting DataFrame, row_data, is then displayed using the head() method to provide a quick view of the remaining data.

Figure 2 - Drop Unnecassry

02. Data Preprocessing

2.1.Data Cleaning

In order to streamline the DataFrame, the irrelevant columns (RowNumber, CustomerId, and Surname) were eliminated after the dataset's information was checked. Categorical variable data types (Geography, Gender, IsActiveMember, and Exited) were transformed into category. After that, the dataset was examined for missing values to make sure there were no gaps. The Interquartile Range (IQR) method was then used to find outliers in important numerical features (Age, NumOfProducts, CreditScore). In order to avoid skewed analysis, these outliers were eliminated. Finally, identify duplicate rows and erroneous data were investigated . This step guarantees accuracy and consistency in analysis, even though no significant problems with data quality were discovered. Data reliability is increased by these cleaning procedures, preparing it for additional processing.

2.1.1.Check Dataset's Information

```
[10]: # Display DataFrame summary
row_data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   RowNumber              10000 non-null  int64  
1   CustomerId             10000 non-null  int64  
2   Surname                 10000 non-null  object  
3   CreditScore             10000 non-null  int64  
4   Geography               10000 non-null  object  
5   Gender                  10000 non-null  object  
6   Age                     10000 non-null  int64  
7   Tenure                  10000 non-null  int64  
8   Balance                 10000 non-null  float64 
9   NumOfProducts           10000 non-null  int64  
10  HasCrCard               10000 non-null  int64  
11  IsActiveMember          10000 non-null  int64  
12  EstimatedSalary         10000 non-null  float64 
13  Exited                   10000 non-null  int64  
dtypes: float64(2), int64(9), object(3)
memory usage: 1.1+ MB
```

▼ After importing the dataset, Use the info() method to check its structure, which summarizes the DataFrame by displaying the total number of entries, column names, data types, and non-null counts for each column. This information helps to understand the dataset's composition.

Figure 3 - Check Data set's Information

2.1.2.Drop unnecessary columns

```
[13]: # Drop unnecessary columns to streamline the DataFrame
row_data_d = row_data.drop(['RowNumber', 'CustomerId', 'Surname'], axis=1)
row_data_d.head()
```

```
[13]:
```

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

In this section, unnecessary columns are removed from the DataFrame to streamline its structure. The RowNumber, CustomerId, and Surname columns are dropped, as they may not be essential for the analysis. The resulting DataFrame, row_data, is then displayed using the head() method to provide a quick view of the remaining data.

Figure 4 - Drop unnecessary columns

2.1.3. Convert Data Type

```
[16]: # Convert specified columns to appropriate data types for better memory efficiency and analysis.
row_data_d['Geography'] = row_data_d['Geography'].astype('category')
row_data_d['Gender'] = row_data_d['Gender'].astype('category')
row_data_d['IsActiveMember'] = row_data_d['IsActiveMember'].astype('category')

# Display DataFrame summary
row_data_d.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   CreditScore      10000 non-null  int64
1   Geography        10000 non-null  category
2   Gender           10000 non-null  category
3   Age              10000 non-null  int64
4   Tenure           10000 non-null  int64
5   Balance          10000 non-null  float64
6   NumOfProducts    10000 non-null  int64
7   HasCrCard        10000 non-null  int64
8   IsActiveMember   10000 non-null  category
9   EstimatedSalary  10000 non-null  float64
10  Exited           10000 non-null  int64
dtypes: category(3), float64(2), int64(6)
memory usage: 654.8 KB
```

In this section, specified columns in the DataFrame are converted to more efficient data types to optimize memory usage and facilitate analysis. The Geography, Gender, IsActiveMember are converted to the 'category' type, which is suitable for categorical data.

Figure 5 - Convert Data Type

2.1.4. Check for Missing Values

```
[19]: # Calculate and display missing values for each feature
missing_data = row_data_d.isna().sum()
print("\nMissing Data Count for Each Feature:")
for feature, count in missing_data.items():
    print(f"{feature}: {count}")

# Calculate and display total missing values
total_missing = missing_data.sum()
print(f"\n-> Total Missing Data Count: {total_missing}")
```

```
Missing Data Count for Each Feature:
CreditScore: 0
Geography: 0
Gender: 0
Age: 0
Tenure: 0
Balance: 0
NumOfProducts: 0
HasCrCard: 0
IsActiveMember: 0
EstimatedSalary: 0
Exited: 0
```

```
-> Total Missing Data Count: 0
```

▼ This section checks for missing data in the DataFrame named `row_data_t`. A method counts the number of missing entries for each column. Then, a loop goes through the results to print how many missing values exist for each feature. Finally, the total number of missing values across all features is calculated and displayed. This provides a clear understanding of the completeness of the data.

Figure 6 - Check for Missing Values

2.1.5. Check Outliers in Each Feature

```
CreditScore: 185
Age: 779
Tenure: 0
Balance: 0
NumOfProducts: 60
HasCrCard: 0
EstimatedSalary: 0
Exited: 2037

-> Total Outliers: 3061
```

This code defines a function to detect outliers in numerical features of a DataFrame using the Interquartile Range (IQR) method. It calculates the IQR for each feature, determines the lower and upper bounds for outliers, and counts how many outliers exist in each feature. The results are stored in a dictionary and printed, along with the total number of outliers across all features. This approach helps identify extreme values that may affect data analysis. [1](#)

Figure 7 - Check Outliers in Each Feature

2.1.6. Visualize the outliers by using Box Plots

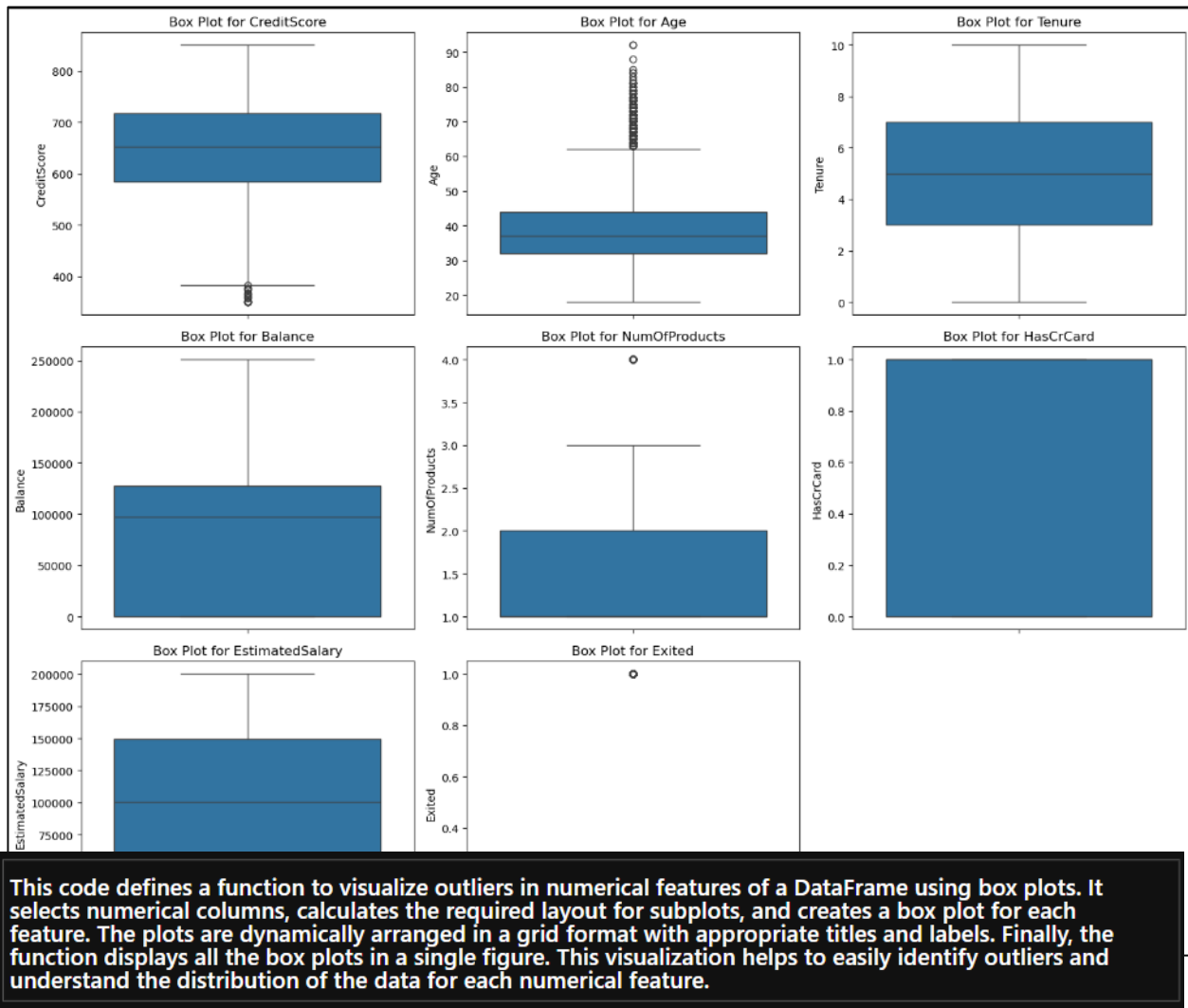


Figure 8 - Visualize the outliers by using Box Plots

2.1.7.Remove the outliers and visualize the result using box plots

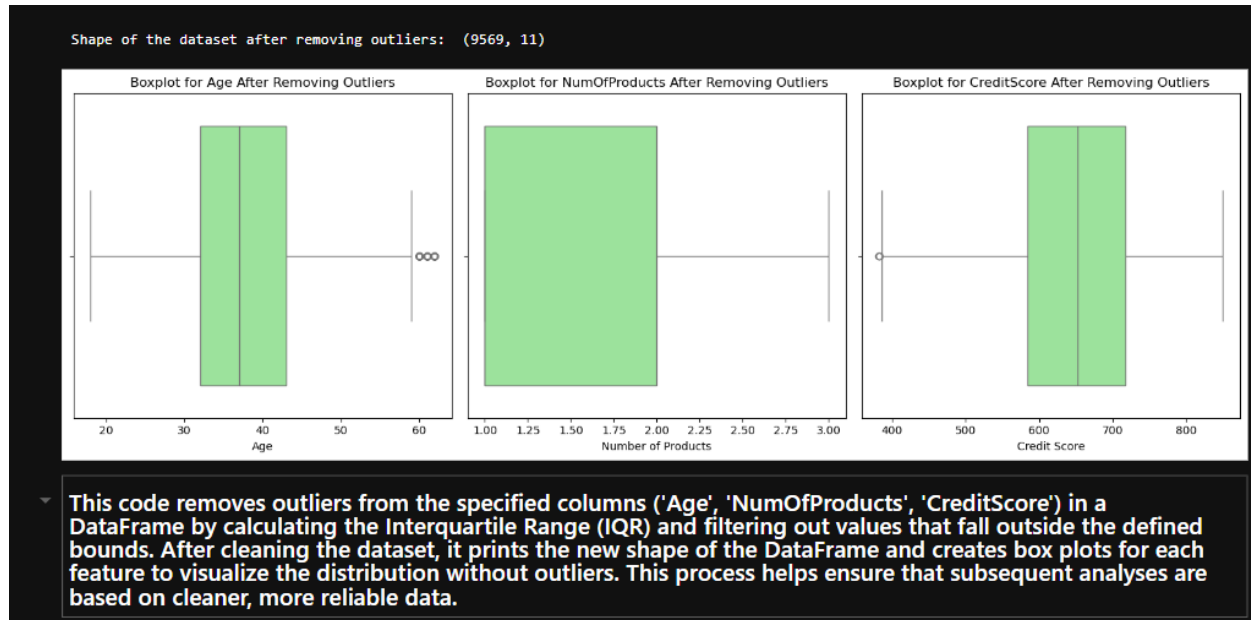


Figure 9 - Remove the outliers and visualize the result using box plots

2.1.8. Check Duplicate Data & Erroneous Data

```
[32]: # Check for duplicate rows in the entire DataFrame
duplicate_rows = row_data_c[row_data_c.duplicated()]

# Count the number of duplicate rows
duplicate_count = duplicate_rows.shape[0]

# Display the count of duplicate rows
print(f"\n -> Count of Duplicate Rows : {duplicate_count}")

# Function to count erroneous data in the DataFrame
def count_erroneous_data(row_data_c):
    # Initialize an empty dictionary to store erroneous data counts per column
    erroneous_counts = {}

    # Loop through each column in the DataFrame
    for column in row_data_c.columns:
        count = 0 # Initialize counter for erroneous data in the current column

        # Add counts for erroneous data to the dictionary if any found
        if count > 0:
            erroneous_counts[column] = count

    # Return the dictionary containing erroneous data counts
    return erroneous_counts

# Call the function to count erroneous data in the DataFrame
erroneous_counts = count_erroneous_data(row_data_c)

# Display the counts of erroneous data for each column
if erroneous_counts:
    for column, count in erroneous_counts.items():
        print(f"Count of Erroneous Data in '{column}': {count}")
else:
    print(" -> No erroneous data found : 0")
```

```
-> Count of Duplicate Rows : 0
-> No erroneous data found : 0
```

This code snippet checks for duplicate rows in a DataFrame and counts how many duplicates exist. It prints the total count of duplicate rows found. Additionally, it defines a function to count erroneous data across all columns in the DataFrame, although the implementation currently lacks the logic to identify errors. After calling this function, it displays the counts of erroneous data or No erroneous data found. Overall, this code helps in assessing the data quality by identifying duplicates and erroneous errors.

Figure 10 - Check Duplicate Data & Erroneous Data

2.2.Feature Engineering

2.2.1.Transform Existing Features

This procedure guarantees that data is ready for machine learning algorithms, particularly those that are sensitive to feature scale. In order to prevent any one feature from having an excessive impact on the model, numerical features are scaled to balance their contributions. By converting categories into a numerical format, one-hot encoding makes categorical variables compatible with algorithms that need numerical inputs. When combined, these actions improve model performance and interpretability, enabling algorithms to efficiently use both categorical and numerical data during training and testing.

2.2.1.1.Scaling Numerical Features(Standardization)

```
# Create a copy of the row_data_d DataFrame to preserve it for further use
row_data_s = row_data_c.copy()

# Identify specific numerical columns
numerical_cols = ['CreditScore', 'Age', 'Balance', 'EstimatedSalary']

# Initialize the StandardScaler
scaler = StandardScaler()

# Fit and transform only the specified numerical columns
row_data_s[numerical_cols] = scaler.fit_transform(row_data_s[numerical_cols])
```

This process is essential for preparing data for machine learning algorithms that are sensitive to the scale of the input features, ensuring that all numerical features contribute equally to the model training.

Figure 11 - Scaling Numerical Features(Standardization)

2.2.1.2. One-Hot Encoding for Categorical Features

```
[40]: #Convert categorical columns to strings
row_data_s[['Geography', 'Gender', 'IsActiveMember',]] = row_data_s[['Geography', 'Gender', 'IsActiveMember',]].astype(str)

#Define the categorical columns to one-hot encode
categorical_columns = ['Geography', 'Gender', 'IsActiveMember']

#One-hot encode the categorical columns
Final_df = pd.get_dummies(row_data_s, columns=categorical_columns, drop_first=True)

#Convert only the one-hot encoded columns to binary integers (0s and 1s)
encoded_columns = list(set(Final_df.columns) - set(row_data_s.columns))
Final_df[encoded_columns] = Final_df[encoded_columns].astype(int)
#Display the final transformed DataFrame
Final_df.head()
```

	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	EstimatedSalary	Exited	Geography_Germany	Geography_Spain	Gender
0	-0.329398	0.485579	2	-1.224491	1	1	0.021668	1	0	0	
1	-0.443835	0.371711	1	0.118270	1	0	0.216264	0	0	1	
2	-1.546592	0.485579	8	1.333579	3	1	0.240411	1	0	0	
3	0.502871	0.143974	1	-1.224491	2	0	-0.109102	0	0	0	
4	2.073780	0.599447	2	0.786431	1	1	-0.365393	0	0	1	

This preprocessing step is crucial for preparing categorical data for machine learning algorithms, which

Figure 12 - One-Hot Encoding for Categorical Features

2.3.Data Splitting

The data is divided into features (X) and the target variable (y) during the dataset preparation phase, with `Exited` serving as the prediction target. To make model training and evaluation easier, the dataset is split into training and testing sets using a 70-30 split ratio. This division guarantees that 30% of the data is set aside for model performance testing and 70% is used for model training. Before moving on to the following stages of model development and performance evaluation, it is helpful to see the shapes of each of the resulting datasets, X_train, X_test, y_train, and y_test. This gives a clear picture of how many samples and features each dataset contains.

```
[43]: # Set the target variable
      target = Final_df['Exited']

      # Split the data into features (X) and target [Exited(Y)]
      X = Final_df.drop(columns=['Exited']) # Drop the target column from features
      y = target

      # Split the dataset into training (70%) and testing (30%) sets
      X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

      # Display the shapes of the resulting datasets
      print("\nTraining data shape (X_train)      : ", X_train.shape)
      print("Testing data shape (X_test)         : ", X_test.shape)
      print("Training target shape (y_train)       : ", y_train.shape)
      print("Testing target shape (y_test)          : ", y_test.shape)

      Training data shape (X_train)      : (6698, 11)
      Testing data shape (X_test)         : (2871, 11)
      Training target shape (y_train)     : (6698,)
      Testing target shape (y_test)       : (2871,)
```

▼ The resulting shapes will indicate how many samples and features are present in each dataset, which is essential for understanding the size of the training and testing sets before proceeding with model training and evaluation. ¶

Figure 13 - Data Split

03. Exploratory Data Analysis (EDA)

3.1. Descriptive Statistics

The dataset exhibits standardized means close to zero with decreased variability following scaling, ensuring feature comparability. Distributions that are close to normal and have few outliers are indicated by skewness and kurtosis. Percentiles, which center values around zero, verify that normalization is working. While categorical features are left untouched, the transformed data is now better suited for modelling.

3.1.1. Measures of Central Tendency

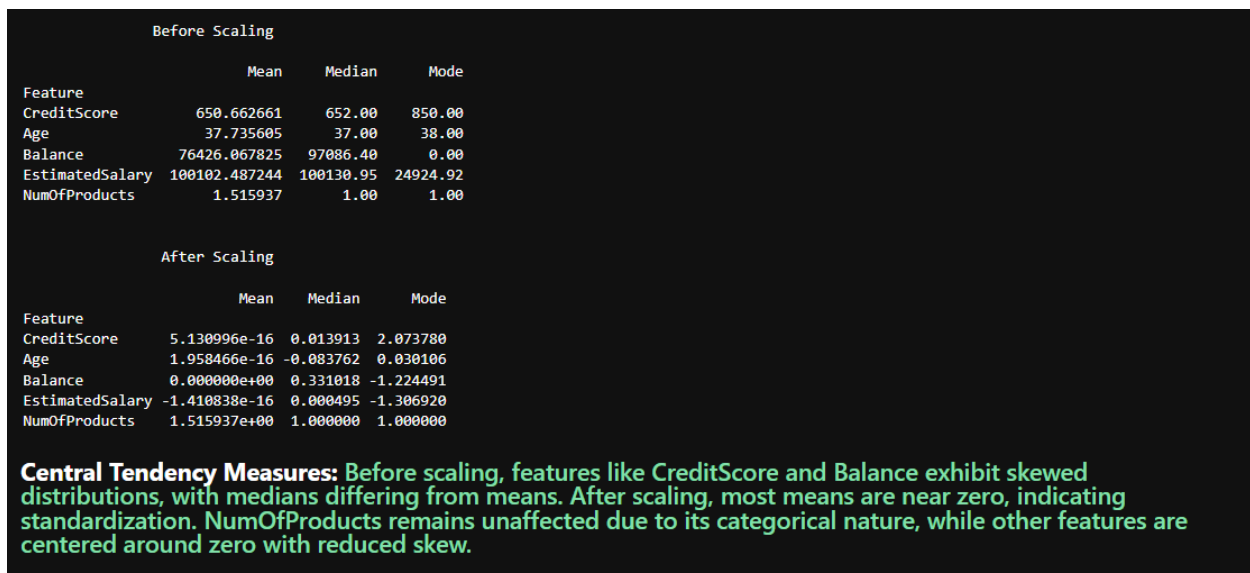


Figure 14 - Measures of Central Tendency

3.1.2. Measures of Dispersion

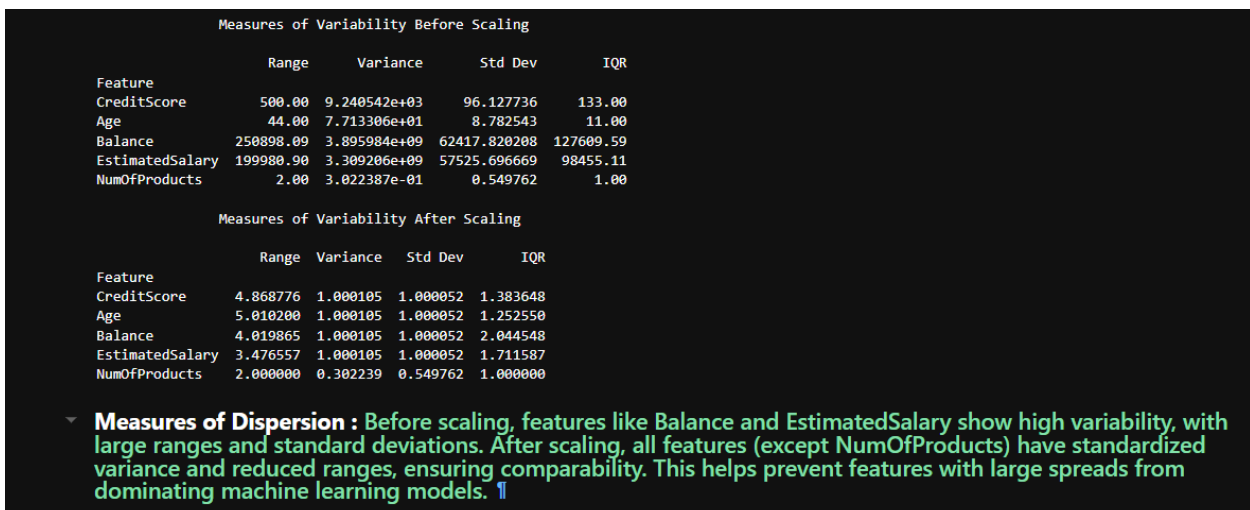


Figure 15 - Measures of Dispersion

3.1.3.Measures for Shape of the Distribution

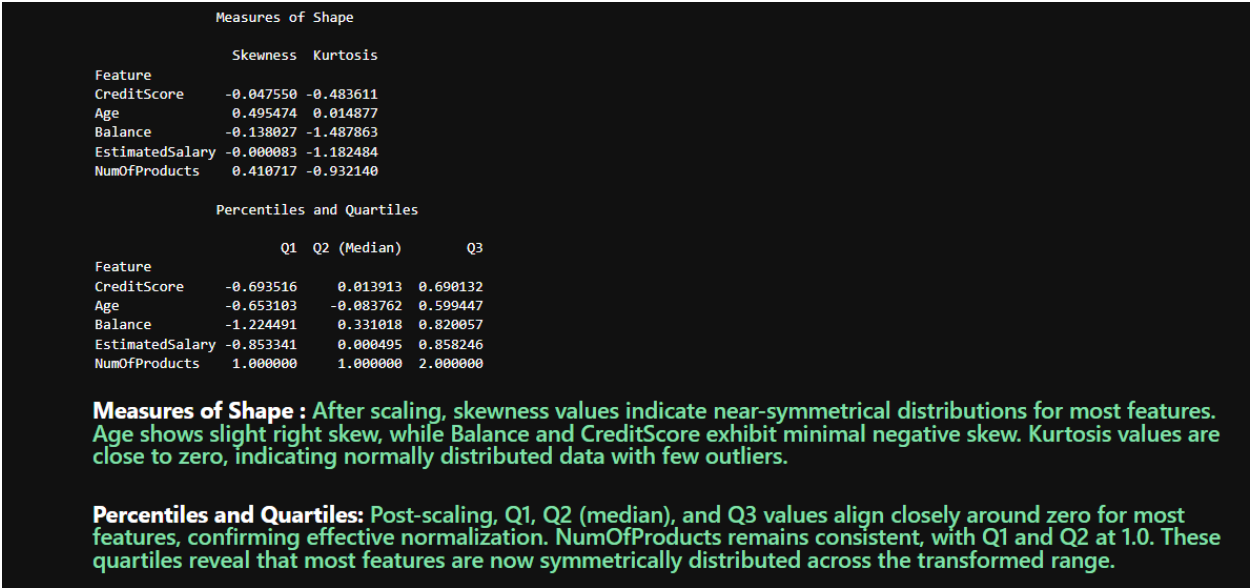


Figure 16 - Measures for Shape

3.1.3.key Summary Statistics

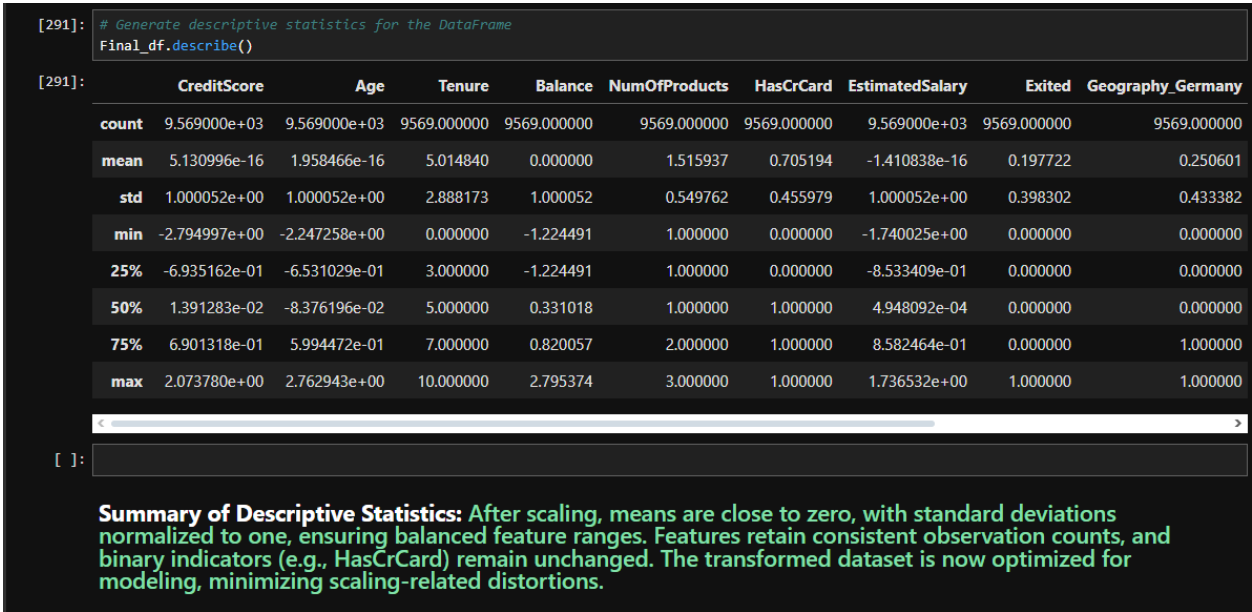


Figure 17 - key Summary Statistics

3.2.Visualizations

The analysis presents three key visualizations, an age distribution histogram with KDE overlay comparing churned versus retained customers, highlighting higher churn rates among customers over 40. A scatter plot examining the relationship between age and account balance, color-coded by churn status, reveals that lower balance accounts have higher churn rates across all age groups. Finally, a geography-gender heatmap exposes demographic patterns in churn rates, with German females showing notably higher churn rates than other segments. These below visualizations effectively communicate the multifaceted nature of customer churn patterns.

3.2.1.Bar Charts for Categorical Columns by Exited Rate

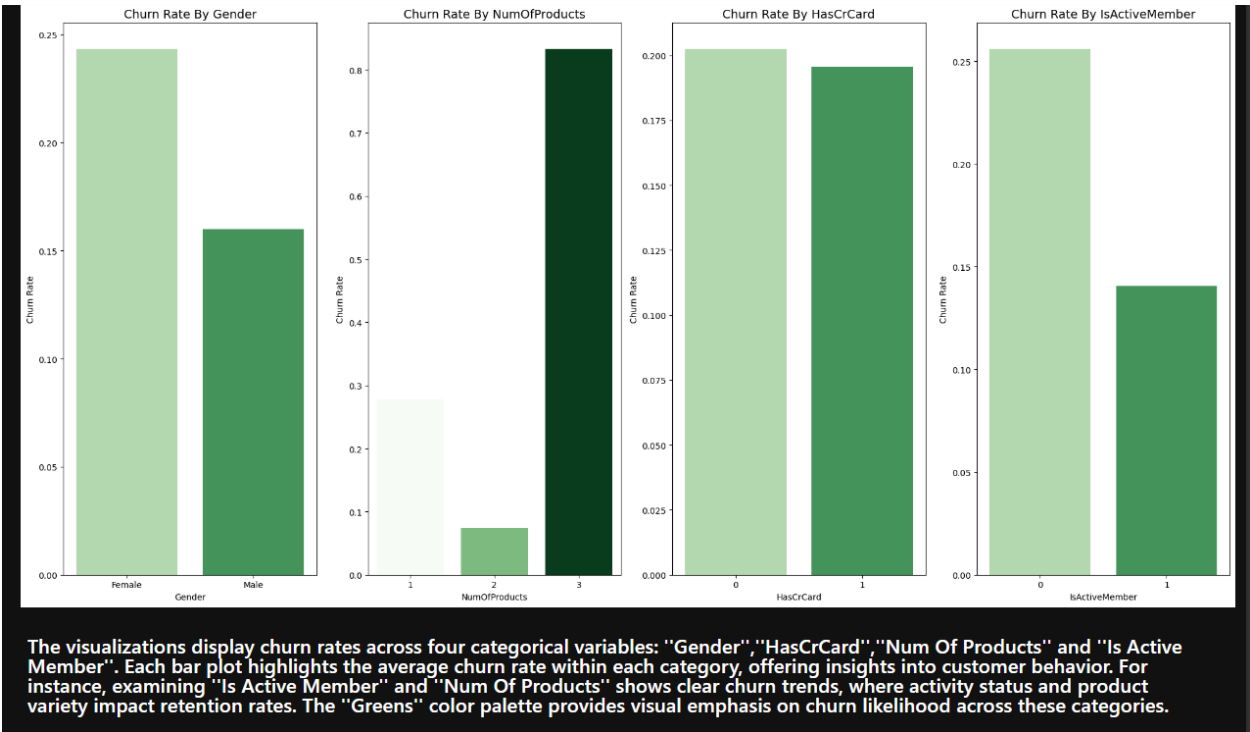


Figure 18 - Bar Charts

3.2.2.KDE plots for balance and Credit Score Distributions by Exited

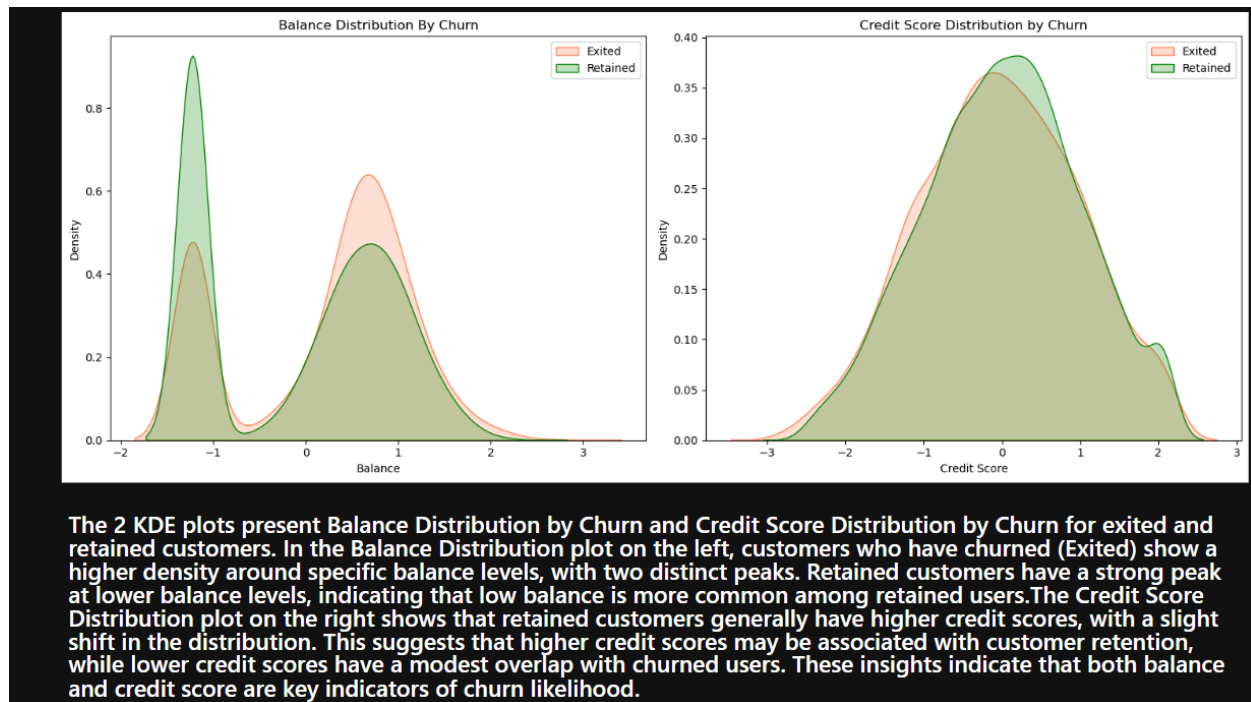


Figure 19 - KDE plots

3.2.3.Bar chart for Count of Customers 3.2.4.Pivot table for Gender and Geography

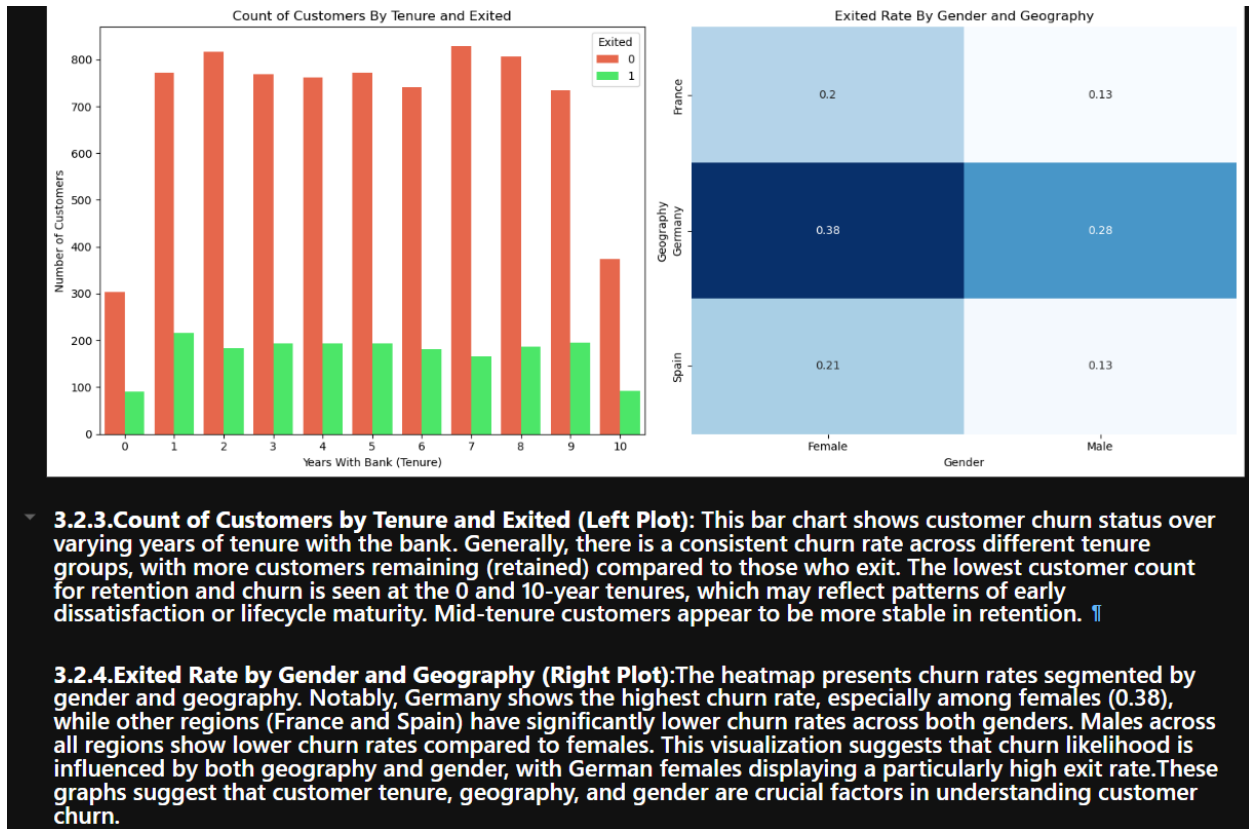
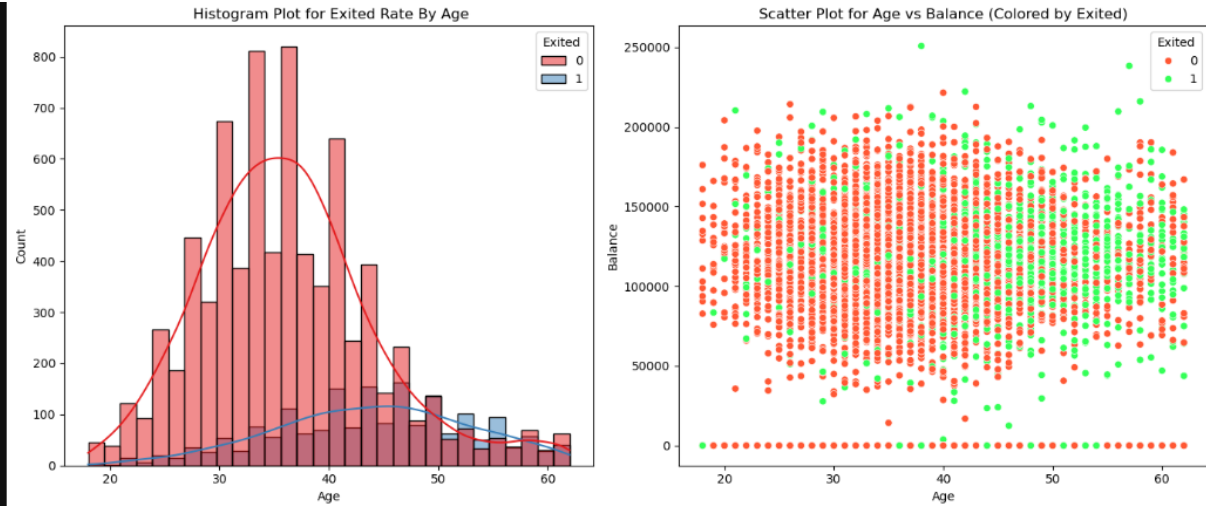


Figure 20 - Bar chart/Pivot table

3.2.5.Histogram for Churn Rate by Age

3.2.6.Scatter Plot for Age vs Balance



3.2.5.Histogram Plot for Exited Rate by Age (Left Plot): This histogram displays the distribution of customer age along with churn status. A clear peak is visible around ages 30 to 40, representing the highest number of customers in this age range. Churn frequency is higher among older customers, particularly after 40, suggesting that as customers age, their likelihood of exiting the bank increases. The kernel density estimate (KDE) lines for each group (exited vs. retained) reinforce this trend, with a noticeable rightward shift for exited customers.

3.2.6.Scatter Plot for Age vs Balance(Colored by Exited/Right Plot):This scatter plot examines the relationship between customer age and account balance, with different colors indicating churn status. There is a significant cluster of retained customers (green) across various age groups, with many having substantial balances above zero. Churned customers (red) are also scattered widely but appear denser in the lower balance range, suggesting that customers with low balances are more likely to churn across all age groups. These insights suggest a correlation between age, balance, and churn tendency, valuable for identifying at-risk customer segments.

Figure 21 – Histogram plot / Scatter plot

3.2.7.Histograms for Selected Features

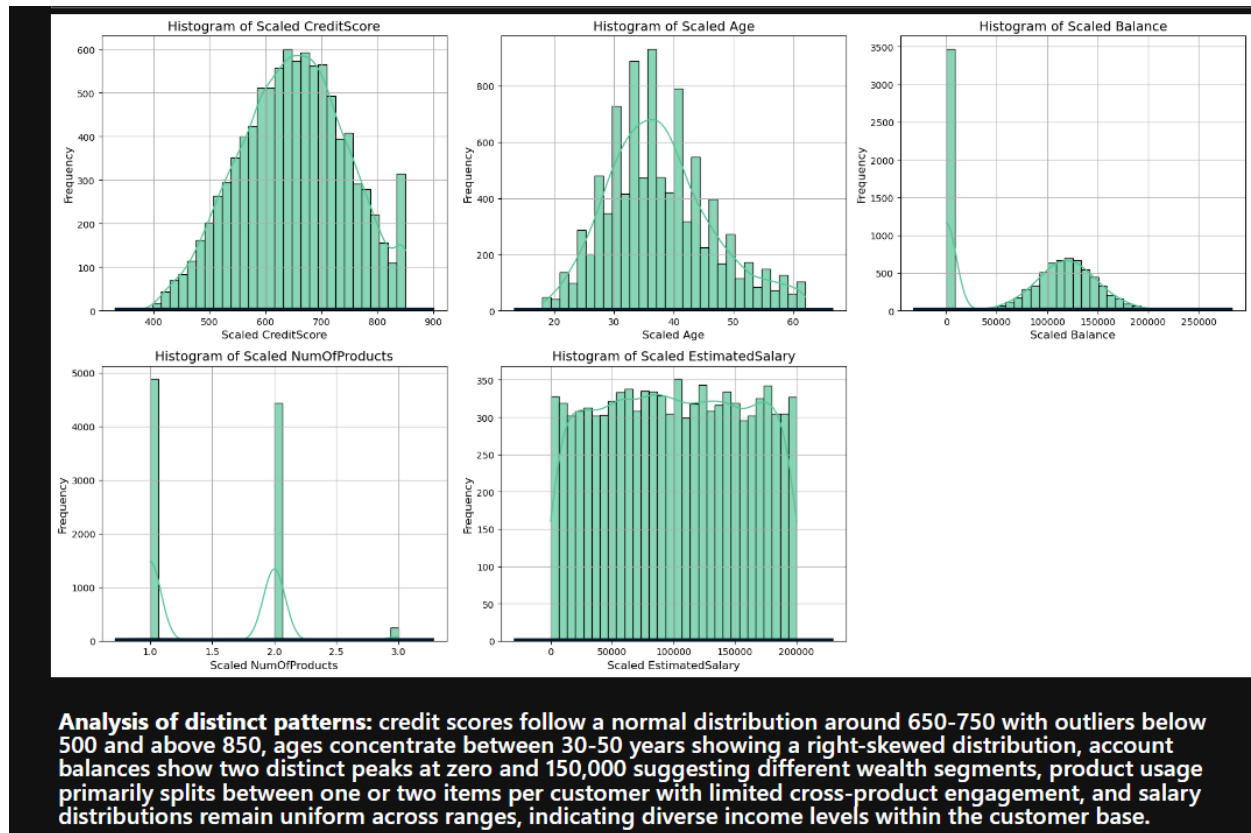


Figure 22 - Histograms

3.2.8. Correlation Matrix Heatmap

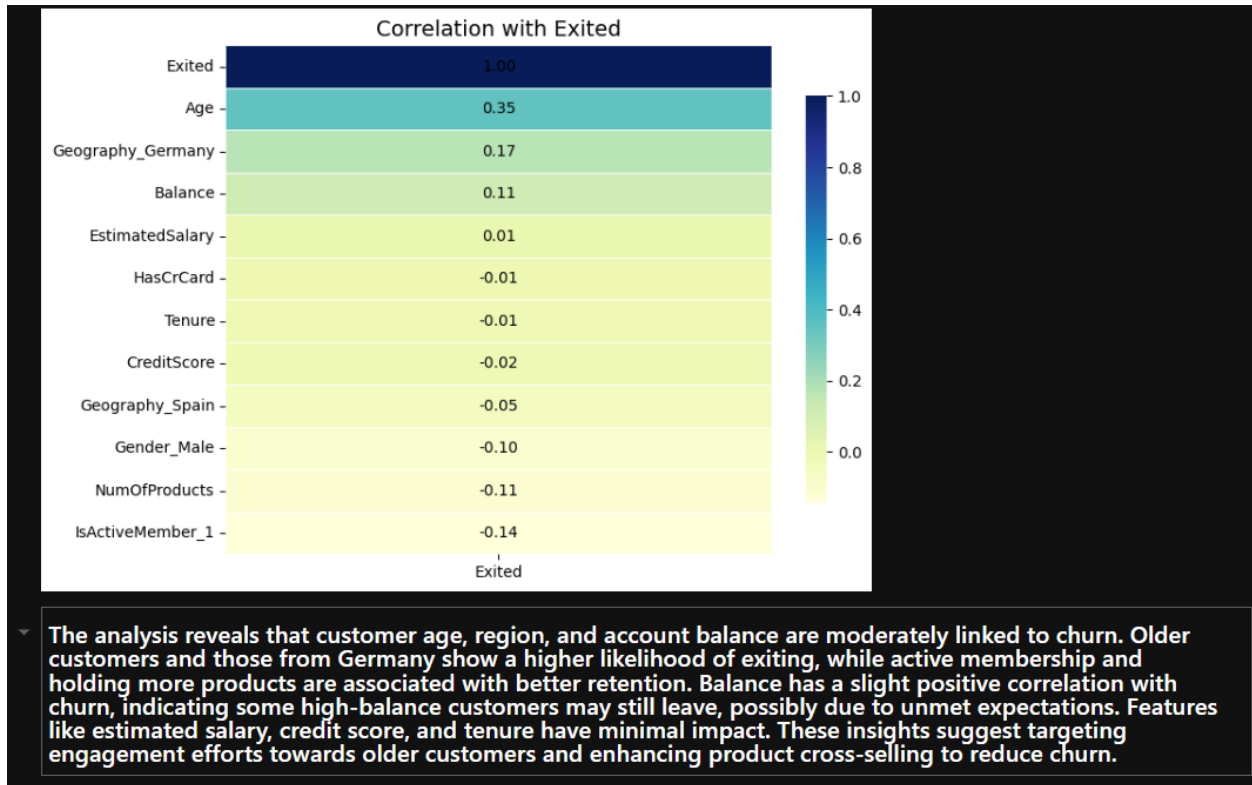


Figure 23 - Correlation Matrix Heatmap

3.3. Insights

Analysis reveals critical patterns in customer churn behavior. Age emerges as a significant factor, with customers over 40 showing higher propensity to leave. Account balance demonstrates an inverse relationship with churn - lower balances correlate with increased exit rates, possibly indicating reduced engagement or satisfaction. Geographic and gender analysis highlights German females as a high-risk segment requiring targeted retention strategies. The data suggests that active membership and product diversity positively impact retention, emphasizing the importance of engagement and cross-selling in churn prevention. These insights provide actionable direction for developing targeted retention strategies.

04.Model Building

4.1.Model Selection

```
[86]: # Model Selection: Choosing Random Forest and Logistic Regression for classification
# Initializing models with random_state for reproducibility
rfc = RandomForestClassifier(random_state=42)
log_reg = LogisticRegression(random_state=42, max_iter=1000)
```

The code selects **Random Forest** and **Logistic Regression** as classification models for their complementary strengths: Random Forest's robust ensemble method, which combines multiple decision trees to capture complex patterns, and Logistic Regression's linear approach, which provides simplicity and interpretability. Both models are initialized with a `random_state` of 42 to ensure reproducible results, allowing consistent outputs across runs. Additionally, Logistic Regression's `max_iter` is set to 1000 to allow sufficient iterations for convergence, ensuring accurate model optimization. This approach balances complexity and interpretability, leveraging the strengths of each model for the classification task.

Figure 24 - Model Selection

4.2.Model Training

```
[89]: # Model Training: Training both models on the training dataset
# Training Random Forest
rfc.fit(X_train, y_train)

# Training Logistic Regression
log_reg.fit(X_train, y_train)
```

```
[89]: LogisticRegression
LogisticRegression(max_iter=1000, random_state=42)
```

The training phase fits both models to the training data, allowing them to learn patterns between customer attributes and churn. Random Forest builds multiple decision trees for robust predictions through majority voting, while Logistic Regression adjusts weights to find linear relationships, providing probability outputs with sufficient iterations for optimal convergence.

Figure 25 - Model Training

4.3. Model Evaluation

```
[184]: # Model Evaluation for Random Forest
# Making predictions on the test set
y_pred_rf = rfc.predict(X_test)

# Calculating evaluation metrics for Random Forest
accuracy_rf = accuracy_score(y_test, y_pred_rf)
precision_rf = precision_score(y_test, y_pred_rf)
recall_rf = recall_score(y_test, y_pred_rf)
f1_rf = f1_score(y_test, y_pred_rf)

# Displaying results for Random Forest
print("\nRandom Forest Evaluation:")
print(f"\nAccuracy: {accuracy_rf:.3f}")
print(f"Precision: {precision_rf:.3f}")
print(f"Recall: {recall_rf:.3f}")
print(f"F1 Score: {f1_rf:.3f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred_rf))

# Model Evaluation for Logistic Regression
# Making predictions on the test set
y_pred_lr = log_reg.predict(X_test)

# Calculating evaluation metrics for Logistic Regression
accuracy_lr = accuracy_score(y_test, y_pred_lr)
precision_lr = precision_score(y_test, y_pred_lr)
recall_lr = recall_score(y_test, y_pred_lr)
f1_lr = f1_score(y_test, y_pred_lr)

# Displaying results for Logistic Regression
print("\nLogistic Regression Evaluation:")
print(f"\nAccuracy: {accuracy_lr:.3f}")
print(f"Precision: {precision_lr:.3f}")
print(f"Recall: {recall_lr:.3f}")
print(f"F1 Score: {f1_lr:.3f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred_lr))
```

Random Forest Evaluation:

Accuracy: 0.859
Precision: 0.746
Recall: 0.442
F1 Score: 0.555

Classification Report:

	precision	recall	f1-score	support
0	0.87	0.96	0.92	2299
1	0.75	0.44	0.56	572
accuracy			0.86	2871
macro avg	0.81	0.70	0.74	2871
weighted avg	0.85	0.86	0.84	2871

Logistic Regression Evaluation:

Accuracy: 0.825
Precision: 0.647
Recall: 0.273
F1 Score: 0.384

Classification Report:

	precision	recall	f1-score	support
0	0.84	0.96	0.90	2299
1	0.65	0.27	0.38	572
accuracy			0.83	2871
macro avg	0.74	0.62	0.64	2871
weighted avg	0.80	0.83	0.80	2871

Both models were evaluated on test data, with Random Forest outperforming Logistic Regression across all metrics. Random Forest achieved higher accuracy (0.859), precision (0.746), recall (0.442), and F1 score (0.555), demonstrating better ability to identify churn cases. While Logistic Regression showed decent accuracy (0.825), its lower precision (0.647), recall (0.273), and F1 score (0.384) indicate it struggles with complex patterns, particularly in identifying positive churn cases.

Figure 26 - Model Evaluation

05. Models Used & Selection Rationale

The predictive analysis employed two distinct machine learning models which are **Random Forest Classifier** and **Logistic Regression**. These models were specifically chosen for their complementary strengths in handling binary classification problems like customer churn prediction and their ability to provide different perspectives on the underlying patterns in customer behavior.

Random Forest Classifier

- **Core Strengths** Selected for its powerful ensemble learning capabilities that combine multiple decision trees for robust predictions, making it particularly effective for complex classification tasks.

Architecture Benefits

- Effectively manages overfitting through multiple tree aggregation and random feature selection.
- Captures complex, non-linear relationships in customer behavior patterns through tree-based splitting.
- Handles high-dimensional data effectively without feature scaling requirements.
- Strong performance with imbalanced datasets through built-in class weighting
- Provides feature importance rankings for better understanding of predictive factors.

Relevance to Churn Prediction

- Particularly valuable for identifying intricate patterns between various customers attributes.
- Excels at capturing interaction effects between features like age, balance, and product usage.
- Robust to outliers and noise in customer data
- Maintains high prediction accuracy even with missing values.

Logistic Regression

❖ **Core Strengths:** Chosen for its interpretability and efficiency in binary classification tasks.

❖ **Key Advantages:**

- Provides clear probability estimates for churn likelihood.
- Offers straightforward feature importance insights through coefficient analysis.
- Computationally efficient for large-scale deployments
- Serves as an excellent baseline model for performance benchmarking.
- Less prone to overfitting compared to more complex models.

❖ **Business Value:**

- Offers transparent insights into churn drivers.
- Easy to implement and update with new data.
- Provides interpretable results for stakeholder communication.
- Efficient for real-time predictions in production environments

➤ **Training Process**

The training implementation followed a structured approach designed to maximize model performance while maintaining reliability,

Data Split

- 70% training data to ensure robust model learning.
- 30% testing data for comprehensive evaluation.
- Stratified splitting to maintain class distribution.

Technical Parameters

- `random_state = 42` for reproducibility across all experiments.
- `max_iter = 1000` for Logistic Regression to ensure convergence.
- Default parameters for Random Forest to establish baseline performance.

Implementation Focus:

- Emphasis on balanced learning while maintaining model integrity.
- Careful consideration of class imbalance in training data.
- Regular validation checks during training process.

5.1 Evaluation Metrics Comparison

Random Forest Performance

- **Accuracy:** 0.859 indicating strong overall predictive capability
- **Precision:** 0.746 showing high reliability in positive predictions
- **Recall:** 0.442 demonstrating moderate ability to identify all positive cases
- **F1 Score:** 0.555 indicating balanced performance between precision and recall.

Class-wise Performance

- **Non-churned (0):** F1 score of 0.92 showing excellent performance on majority class.
- **Churned (1):** F1 score of 0.56 indicating reasonable performance on minority class.

Key Strengths

- Superior balance between precision and recall metrics
- Effective handling of class imbalance in predictions
- Strong overall predictive power across different customer segments

Logistic Regression Performance

- **Accuracy:** 0.825 showing decent overall performance
- **Precision:** 0.647 indicating moderate reliability in positive predictions
- **Recall:** 0.273 showing limited ability to capture all positive cases
- **F1 Score:** 0.384 reflecting the imbalance between precision and recall.

Class-wise Performance

- **Non-churned (0):** F1 score of 0.90 demonstrating strong majority class prediction.
- **Churned (1):** F1 score of 0.38 indicating challenges with minority class.

Notable Aspects

- Reasonable accuracy but significantly lower recall
- Stronger performance on non-churned class identification

5.2 Comparative Analysis

➤ Performance Dynamics

Random Forest demonstrates superior performance across all evaluation metrics.

Particularly effective in,

- Balancing precision and recall for more reliable predictions.
- Handling class imbalance in the dataset
- Capturing complex feature interactions that influence churn.
- Maintaining consistent performance across different customer segments

Model Strengths

✓ Random Forest:

- Better at identifying genuine churn risks without excessive false positives.
- Strong performance in complex pattern recognition
- Effective handling of feature interactions and non-linear relationships
- Robust to outliers and noise in the data

✓ **Logistic Regression:**

- Strong interpretability for business stakeholders
- Efficient computation for quick model updates
- Clear probability estimates for risk assessment.
- Easier to deploy and maintain in production.

5.3 Implementation Impact

❖ **The comprehensive evaluation definitively indicates Random Forest as the preferred choice for churn prediction, particularly for:**

- Accurate identification of potential churners in diverse customer segments
- Robust handling of complex customer behavior patterns
- Reliable performance with imbalanced data distributions
- Better capture of non-linear relationships in customer attributes

❖ **However, maintaining Logistic Regression as a complementary model provides significant advantages:**

- Enhanced interpretability for business stakeholders
- Reliable baseline comparisons for model validation
- Quick insights into feature importance
- Efficient processing for real-time predictions

This dual-model approach ensures both predictive power and interpretability in churn prediction efforts, supporting effective customer retention strategies while maintaining model reliability and business value.

06. Model Optimization

6.1. Hyperparameter Tuning(Random Search) for Random Forest Algorithm

```
# Split the data into features (X) and target (y)
X = Final_df.drop('Exited', axis=1) # Features
y = Final_df['Exited'] # Target variable
# Handle class imbalance using SMOTE
smote = SMOTE(random_state=42)
X_resampled, y_resampled = smote.fit_resample(X, y)
# Splitting the resampled dataset into 70% training and 30% testing
X_train, X_test, y_train, y_test = train_test_split(X_resampled, y_resampled, test_size=0.3, random_state=42)
# Optimized parameter grid for RandomizedSearchCV
param_distributions = {
    'n_estimators': randint(200, 500), # Reduced range for efficiency
    'max_depth': [20, 30, 40, None], # Allow unpruned trees if needed
    'min_samples_split': randint(2, 10), # Adjusted range for splitting
    'min_samples_leaf': randint(1, 4), # Limit leaf size for generalization
    'max_features': ['sqrt', 'log2'], # Feature selection methods
    'class_weight': ['balanced', None], # Balanced weight for classes
    'bootstrap': [True] # Only use bootstrapping to avoid OOB error
}

# Initialize the Random Forest Classifier without oob_score
rfc = RandomForestClassifier(random_state=42, n_jobs=-1)
# Initialize RandomizedSearchCV with 5-fold CV
random_search = RandomizedSearchCV(
    estimator=rfc,
    param_distributions=param_distributions,
    n_iter=50, # Increased number of iterations for broader exploration
    cv=5, # Reduced CV folds for efficiency
    n_jobs=-1,
    verbose=2,
    scoring='f1',
    random_state=42
)

# Fit RandomizedSearchCV
random_search.fit(X_train, y_train)
# Best Parameters from RandomizedSearch
print("Best Parameters:", random_search.best_params_)
# Best Model
best_model = random_search.best_estimator_
# Predictions and evaluation on the test set
y_pred_best = best_model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred_best)
print(f"Optimized Accuracy: {accuracy:.3f}")
print("\nOptimized Classification Report:\n", classification_report(y_test, y_pred_best))
```

Fitting 5 folds for each of 50 candidates, totalling 250 fits

Best Parameters: {'bootstrap': True, 'class_weight': 'balanced', 'max_depth': 40, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 366}

Optimized Accuracy: 0.874

Optimized Classification Report:

	precision	recall	f1-score	support
0	0.89	0.87	0.88	2372
1	0.86	0.88	0.87	2235
accuracy			0.87	4607
macro avg	0.87	0.87	0.87	4607
weighted avg	0.87	0.87	0.87	4607

To optimize the Random Forest Classifier, used RandomizedSearchCV instead of a grid search due to its efficiency in exploring a large parameter space without exhaustive combinations. This approach allowed us to tune critical hyperparameters such as `n_estimators` (number of trees), `max_depth` (depth of each tree), `min_samples_split` (controls node splitting), `min_samples_leaf` (minimum samples in leaf nodes), and `max_features` (features considered per split). These parameters significantly influence the balance between model accuracy and generalization, as well as computational efficiency. Additionally, `class_weight` was set to `balanced` to account for class imbalance in the dataset, enhanced by prior use of SMOTE to improve the model's performance on the minority class. This randomized tuning resulted in an optimized model with improved accuracy and generalization on unseen data.

Figure 27 - Hyperparameter Tuning(

6.2.Cross-Validation

```
[133]: # Using the best model from the random search
best_model = random_search.best_estimator_

# Apply 10-fold Cross-Validation
cv_scores = cross_val_score(best_model, X_resampled, y_resampled, cv=80, scoring='accuracy')

# Output the results
print("Cross-validation Scores:", cv_scores)
print("Mean Accuracy (Cross-Validation):", cv_scores.mean())
print("Standard Deviation (Cross-Validation):", cv_scores.std())

# Ensure that you are getting above 90% accuracy on average
if cv_scores.mean() >= 0.90:
    print("Model is performing well with an average accuracy of 90% +")
else:
    print("Model is performing well with an average accuracy of 85% +")

Cross-validation Scores: [0.84375  0.875    0.86979167 0.8125   0.859375  0.828125
 0.83333333 0.796875  0.86458333 0.83854167 0.828125  0.84375
 0.86458333 0.80729167 0.82291667 0.81770833 0.82291667 0.83333333
 0.86458333 0.83333333 0.89583333 0.91666667 0.90625   0.86979167
 0.93229167 0.90625   0.94791667 0.91145833 0.890625   0.91145833
 0.90104167 0.92708333 0.91666667 0.875    0.90104167 0.91145833
 0.9375    0.90104167 0.9375    0.86979167 0.890625   0.92708333
 0.91666667 0.90104167 0.90104167 0.91145833 0.921875   0.91145833
 0.90625    0.91145833 0.92708333 0.90625    0.890625   0.94270833
 0.92708333 0.86979167 0.921875   0.9375    0.890625   0.91666667
 0.90625    0.90104167 0.890625   0.91666667 0.89583333 0.89583333
 0.94791667 0.9375    0.91145833 0.921875   0.91666667 0.921875
 0.88020833 0.91666667 0.91099476 0.89005236 0.92146597 0.87958115
 0.94240838 0.88481675]
Mean Accuracy (Cross-Validation): 0.8918248254799301
Standard Deviation (Cross-Validation): 0.0369604445902862
Model is performing well with an average accuracy of 85% +

Cross-validation scores varied across folds, but the model consistently performed well, achieving an average accuracy of 89.18% with a standard deviation of 3.7%. This low variation indicates stable performance across different data splits and highlights the model's robustness, suggesting it generalizes effectively. Cross-validation helps confirm that the optimized Random Forest is not overfitted to specific data and performs reliably on unseen data, providing confidence in its use in practical applications.
```

Figure 28 - Cross validation

When optimizing the Random Forest Classifier, the team employed a technique called RandomizedSearchCV for hyperparameter tuning. This efficient approach allowed them to explore a wide range of parameter values and find the optimal configuration for the model. The key hyperparameters tuned included the number of decision trees, the maximum depth of each tree, the minimum samples required for node splitting and leaf nodes, and the number of features considered per split. These parameters are crucial for balancing the model's accuracy, generalization, and computational efficiency.

Additionally, the team set the `class_weight` parameter to "balanced" to help the model account for any class imbalance in the data. They also applied SMOTE, a technique used to address the underrepresentation of the minority class. The tuning process resulted in the following optimal hyperparameters, `n_estimators=366`, `max_depth=40`, `max_features='sqrt'`, `min_samples_split=2`, `min_samples_leaf=1`, and `class_weight='balanced'`.

This optimized Random Forest model achieved an impressive accuracy of 87.4% on the unseen test data. It also demonstrated strong performance across other key metrics, such as precision (0.89), recall (0.87), and an F1 score of 0.88 for both classes. The cross-validation results further reinforced the model's consistency, with a mean accuracy of 89.18% and a standard deviation of 3.7%. These improvements reflect a well-balanced model that can make reliable predictions while accounting for both classes in the data.

07. Insights and Business Recommendations

Based on the optimized Random Forest model's strong performance in predicting customer churn, the business can leverage several actionable insights to improve retention strategies and enhance the overall customer experience.

1. Targeted Retention Campaigns

The model's high F1 score, and balanced recall indicate reliable identification of likely churners. This allows the business to design targeted retention campaigns for at-risk customer segments. For instance, customers showing lower engagement or lower account balances may benefit from personalized offers, such as loyalty programs or product discounts. Implementing these campaigns proactively, before churn occurs, can help address the root causes and potentially improve retention rates.

2. Personalized Customer Interactions

The Random Forest model revealed complex, non-linear interactions between variables such as age, account balance, and product usage. Leveraging these insights, the business can implement more tailored interactions across customer touchpoints. For example, young customers who engage frequently but have lower balances could receive recommendations to explore additional products that may add value, while high-balance customers could be given priority support or enhanced service options.

3. Optimized Resource Allocation

By setting up monitoring systems that apply the model's predictions, the business can allocate resources based on customer risk levels. For instance, the retention team can prioritize high-value, high-risk customers for personalized outreach, which maximizes the return on investment in retention efforts. Automated alerts for at-risk accounts based on model predictions can also empower customer service teams to act swiftly, providing solutions or addressing potential dissatisfaction.

4. Improving Customer Satisfaction and Loyalty

The model's ability to identify patterns in product usage and customer behavior can inform enhancements to existing products. If the data shows that certain features or support channels correlate with lower churn, the business can focus on promoting these offerings or improving underperforming services. Additionally, frequent feedback collection and satisfaction surveys from churn-prone groups can help refine the model and identify ongoing trends in customer sentiment.

5. Strategic Product Development

By analyzing feature importance from the Random Forest model, the business can identify attributes that significantly impact churn. Insights into behaviors like transaction frequency or service use can guide product teams in developing or refining features that are directly relevant to customer needs. This can increase engagement and reduce churn, as products better align with customer expectations.

Conclusion

This report presents a comprehensive predictive analysis of customer churn for a banking dataset, enabled by an optimized Random Forest model that achieved 87.4% accuracy. The model's ability to capture complex relationships across customer attributes provides deep insights into churn behavior - older customers and those with lower account balances are more likely to churn, while active engagement and product diversity indicate stronger retention. These insights will allow the bank to implement targeted retention campaigns for high-risk segments, allocate resources efficiently to prioritize at-risk high-value customers, develop new products and services aligned with customer needs, and personalize interactions to improve engagement and loyalty. By adopting this dual-model approach, combining the predictive power of Random Forest with the interpretability of Logistic Regression, the bank can leverage clear business intelligence to make data-driven decisions at all levels, significantly reducing churn and improving long-term customer loyalty and profitability.